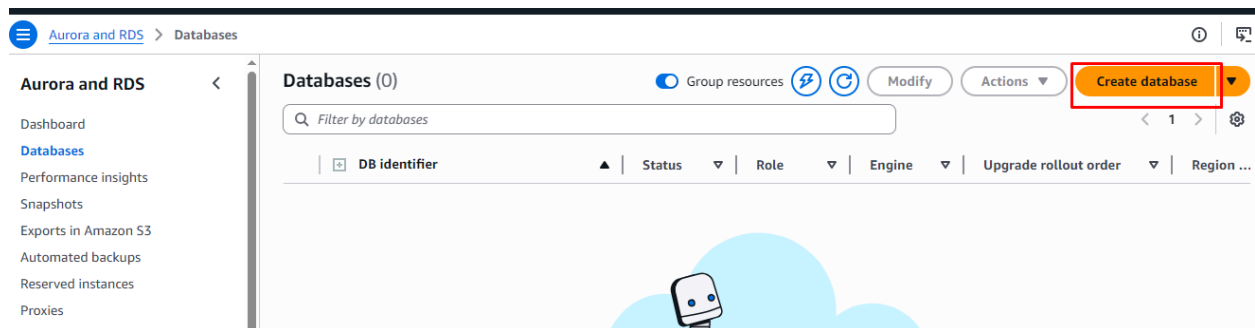
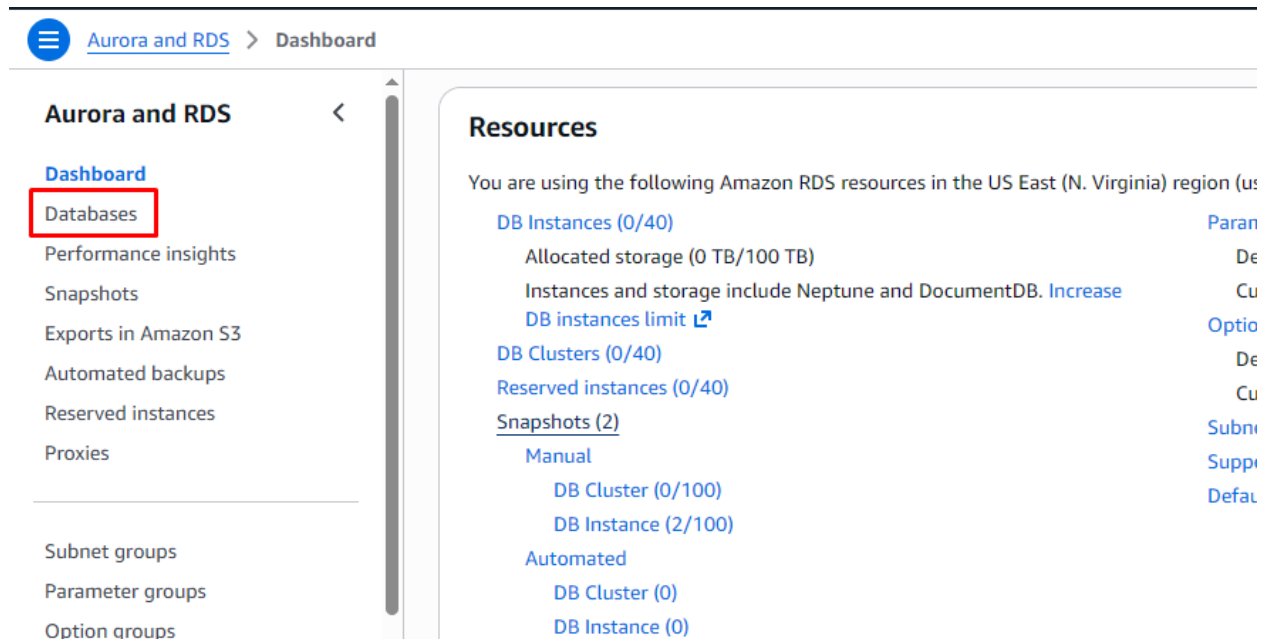
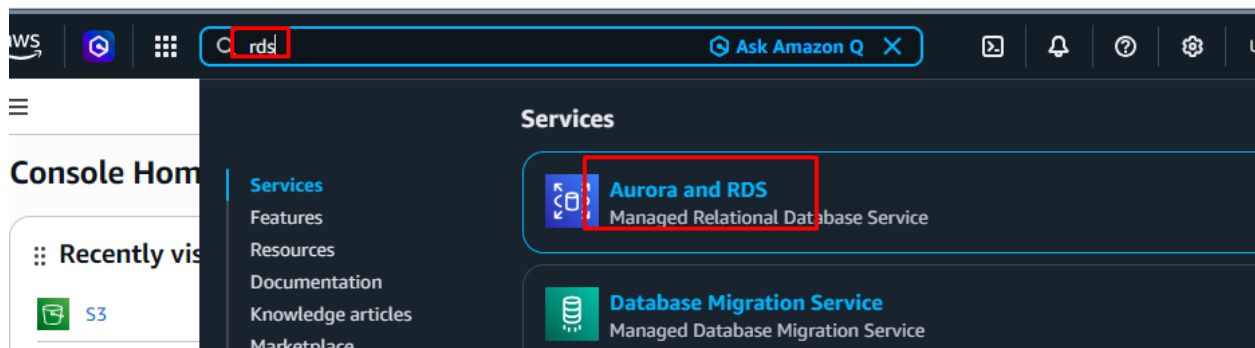


1. Create RDS instance with MYSQL



Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible) 	☐ Aurora (PostgreSQL Compatible) 	☒ MySQL 	☐ PostgreSQL 
☐ MariaDB 	☐ Oracle 	☐ Microsoft SQL Server 	☐ IBM Db2 

Note: Not mandatory to select mysql, based on your comfort you can pick the database software

Choose a database creation method

☒ Full configuration

You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create

Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Templates

Choose a sample template to meet your use case.

☐ Production

Use defaults for high availability and fast, consistent performance.

☐ Dev/Test

This instance is intended for development use outside of a production environment.

☒ Free tier

Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS.

Availability and durability

Availability and durability

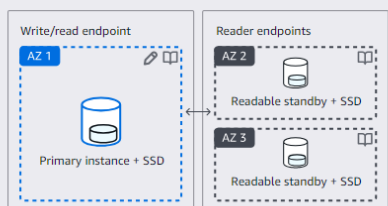
Deployment options [Info](#)

Choose the deployment option that provides the availability and durability needed for your use case. AWS is committed to a certain level of uptime depending on the deployment option you choose. Learn more in the [Amazon service level agreement \(SLA\)](#).

☐ Multi-AZ DB cluster deployment (3 instances)

Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:

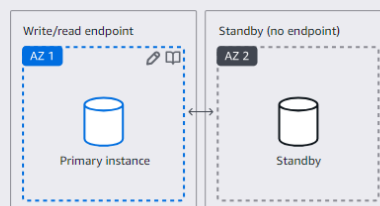
- 99.95% uptime
- Redundancy across Availability Zones
- Increased read capacity
- Reduced write latency



☐ Multi-AZ DB instance deployment (2 instances)

Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:

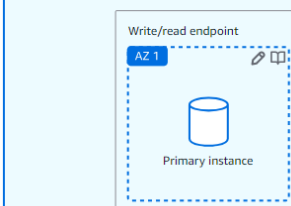
- 99.95% uptime
- Redundancy across Availability Zones



☒ Single-AZ DB instance deployment (1 instance)

Creates a single DB instance without standby instances. This setup provides:

- 99.5% uptime
- No data redundancy



Settings

Edition

☒ MySQL Community

Engine version [Info](#)

View the engine versions that support the following database features.

▼ Hide filters

☐ Show only versions that support the Multi-AZ DB cluster [Info](#)

Create a Multi-AZ DB cluster with one primary DB instance and two readable standby DB instances. Multi-AZ DB clusters provide up to 2x faster transaction commit latency and auto seconds.

☐ Show only versions that support the Amazon RDS Optimized Writes [Info](#)

Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

Engine version

MySQL 8.4.7 ▼

☐ Enable RDS Extended Support [Info](#)

Amazon RDS Extended Support is a [paid offering](#). By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS version. Check the end of standard support date for your major version in the [RDS for MySQL documentation](#).

DB instance identifier [Info](#)

DB instance identifier [Info](#)

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

dvs-genai-b1

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't end with a hyphen.

Credentials Settings

Master username [Info](#)

Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - *most secure*

RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed

Create your own password or have RDS create a password

☐ Auto generate password

Amazon RDS can generate a password for you, or you can specify your own password.

Aurora and RDS

Databases

Create database

Master password

Info

.....

Password strength

Very weak

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' " @

Confirm master password

Info

.....

▼ Additional credentials settings

Database authentication options

Info

☒ Password authentication

Authenticates using database passwords.

☐ Password and IAM database authentication

Authenticates using the database password and user credentials through AWS IAM users and roles.

☐ Password and Kerberos authentication

Choose a directory in which you want to allow authorized users to authenticate with this DB instance using Kerberos Authentication.

Note: Don't give strong password just make it simple like admin1234

Aurora and RDS

Databases

Create database

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class

Info

▼ Hide filters

☐ Show instance classes that support Amazon RDS Optimized Writes

Info

Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

☐ Include previous generation classes

☐ Standard classes (includes m classes)

☐ Memory optimized classes (includes r and x classes)

☒ Burstable classes (includes t classes)

Instance type

db.t3.micro

2 vCPUs 1 GiB RAM

EBS Bandwidth: Up to 2,085 Mbps

Network: Up to 5 Gbps

▼

General Purpose SSD (gp2)

Baseline performance determined by volume size

General Purpose SSD (gp3)

Performance scales independently from storage

Provisioned IOPS SSD (io1)

Flexibility in provisioning I/O

Provisioned IOPS SSD (io2)

Low latency, highly durable, I/O intensive storage

Magnetic

Limited to a maximum of 1,000 IOPS (not recommended)

General Purpose SSD (gp2)

Baseline performance determined by volume size

Storage

Storage type [Info](#)

Provisioned IOPS SSD (io2) storage volumes are now available.

General Purpose SSD (gp2)

Baseline performance determined by volume size

Allocated storage [Info](#)

20

GiB

Allocated storage value must be 20 GiB to 6,144 GiB

► Additional storage configuration

Connectivity [Info](#)

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource is accessible.

☒ Don't connect to an EC2 compute resource

Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ Connect to an EC2 compute resource

Set up a connection to an EC2 compute resource for

Virtual private cloud (VPC) [Info](#)

▼ Additional storage configuration

Storage autoscaling [Info](#)

Provides dynamic scaling support for your database's storage based on your application

☒ Enable storage autoscaling

Enabling this feature will allow the storage to increase after the specified threshold

Maximum storage threshold [Info](#)

Charges will apply when your database autoscales to the specified threshold

Allocated storage value must be 22 GiB to 6,144 GiB

☰ [Aurora and RDS](#) > [Databases](#) > [Create database](#)

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can c

☒ Don't connect to an EC2 compute resource

Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.

☐ Connect to an EC2 compute resource

Set up a connection to an EC2 compute resource for this databas

Virtual private cloud (VPC) [Info](#)

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

Default VPC (vpc-0b8fbad2d38436ee9)

6 Subnets, 6 Availability Zones



After a database is created, you can't change its VPC. Only VPCs with a corresponding DB subnet group are listed.

After a database is created, you can't change its VPC.

DB subnet group [Info](#)

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

default

Public access [Info](#)

☒ Yes

RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to your database using VPC security groups that specify which resources can connect to the database.

Public access [Info](#)

- ☒ **Yes**
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC security groups that specify which resources can connect to the database.
- ☐ **No**
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

VPC security group (firewall) [Info](#)

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ **Choose existing**
Choose existing VPC security groups

☐ **Create new**
Create new VPC security group

Existing VPC security groups

Choose one or more options

default ✕

Availability Zone [Info](#)

No preference

RDS Proxy

RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

☐ **Create an RDS Proxy** [Info](#)

RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy](#).

Certificate authority - optional [Info](#)

Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the certificate authority you provision.

rds-ca-rsa2048-g1 (default)
Expiry: May 26, 2061

If you don't select a certificate authority, RDS chooses one for you.

▼ **Additional configuration**

Database port [Info](#)

TCP/IP port that the database will use for application connections.

3306

Monitoring [Info](#)

Choose monitoring tools for this database. Database Insights provides a combined view of Performance Insights and Enhanced Monitoring for your fleet of databases. Database Insights is separate from RDS monthly estimates. See [Amazon CloudWatch pricing](#).

☐ Database Insights - Advanced

- Retains 15 months of performance history
- Fleet-level monitoring
- Integration with CloudWatch Application Signals

☒ Database Insights - Standard

▼ Additional monitoring settings

Enhanced Monitoring, CloudWatch Logs and DevOps Guru

Enhanced Monitoring

☐ Enable Enhanced monitoring

Enabling Enhanced Monitoring metrics are useful when you want to see how different processes or threads use the CPU.

Log exports

Select the log types to publish to Amazon CloudWatch Logs

- ☐ Audit log
- ☐ Error log
- ☐ General log
- ☐ iam-db-auth-error log

Log exports

Select the log types to publish to Amazon CloudWatch Logs

- ☐ Audit log
- ☐ Error log
- ☐ General log
- ☐ iam-db-auth-error log
- ☐ Slow query log

IAM role

The following service-linked role is used for publishing logs to CloudWatch Logs.

RDS service-linked role

► Additional configuration

~~Database options, encryption turned on,~~ backup turned on, backtrack turned off, maintenance, CloudWatch Logs, del

i You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that :

default.mysql8.4

Option group [Info](#)

default:mysql-8-4

Backup

☒ Enable automated backup

Creates a point-in-time snapshot of your database



Please note that automated backups are currently supported for InnoDB storage engine only. If you are using MyISAM, re

Backup retention period [Info](#)

The number of days (1-35) for which automatic backups are kept.

1



day

Backup window [Info](#)

The daily time range (in UTC) during which RDS takes automated backups.

☐ Choose a window

Ensure you are uncheck the above

default.mysql8.4

Option group [Info](#)

default:mysql-8-4

Backup

☐ Enable automated backup

Creates a point-in-time snapshot of your database

Backup tags [Info](#)

Copies tags from the source database to the automated backup and snapshots respectively.

☐ Copy tags to automated backup

This is a one-time setting. Future tag modifications need manual updates.

☐ Enable encryption

Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console. [Learn more](#)

Maintenance

Auto minor version upgrade [Info](#)

☐ Enable auto minor version upgrade

Enabling auto minor version upgrade will automatically upgrade your database minor version. For limitations and more details, see [Automatically upgrading the minor engine version documentation](#).

Maintenance window [Info](#)

Aurora and RDS > **Databases** > **Create database**

☐ **Enable encryption**
Choose to encrypt the given instance. Master key IDs and aliases appear in the list after they have been created using the AWS Key Management Service console. [Info](#)

Maintenance
Auto minor version upgrade [Info](#)

☐ **Enable auto minor version upgrade**
Enabling auto minor version upgrade will automatically upgrade your database minor version. For limitations and more details, see Automatically upgrading the minor engine version [documentation](#) [↗](#).

Maintenance window [Info](#)
Select the period you want pending modifications or maintenance applied to the database by Amazon RDS.

☐ Choose a window

☒ **No preference**

☐ **Enable deletion protection**
Protects the database from being deleted accidentally. While this option is enabled, you can't delete the database.

① You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.

[Cancel](#) [Create database](#)

Verify:

Aurora and RDS > **Databases** > **dvs-genai-b1**

Connectivity & security | Monitoring | Logs & events | Configuration | Zero-ETL integrations | Maintenance & backups | Data migrations | Tags

Connect using [Info](#)

☐ Code snippets
Use when connecting through SDK, APIs, or third-party tools including agents.

☐ CloudShell
Use for a quick access to AWS CLI that launches directly from the AWS Management Console.

☒ **Endpoints**
Use when connecting through any IDE interface.

Database name
mysql

Master username
admin

Internet access gateway
☐ Disabled

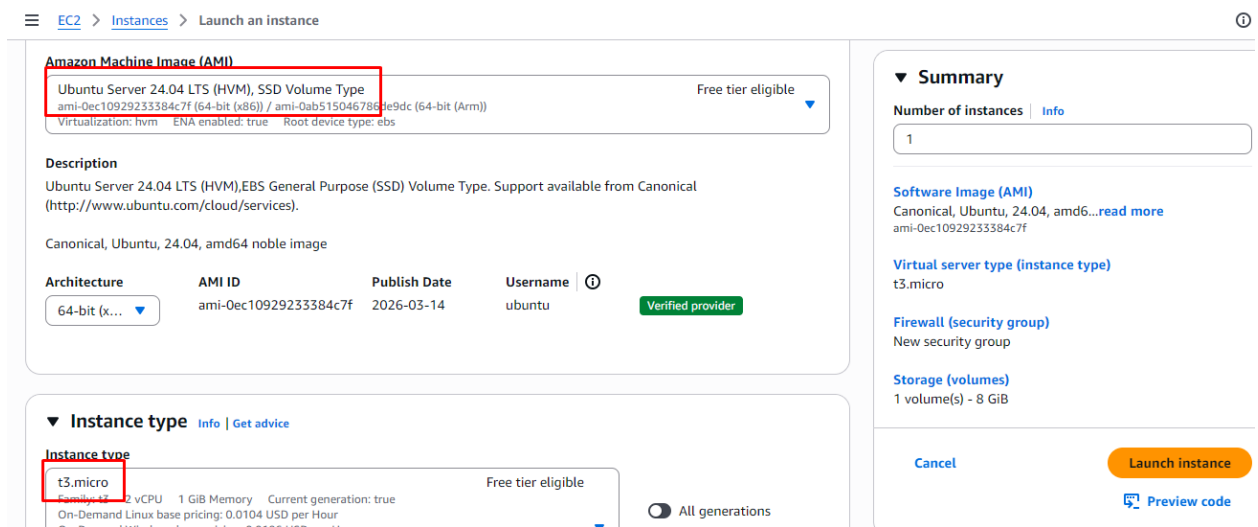
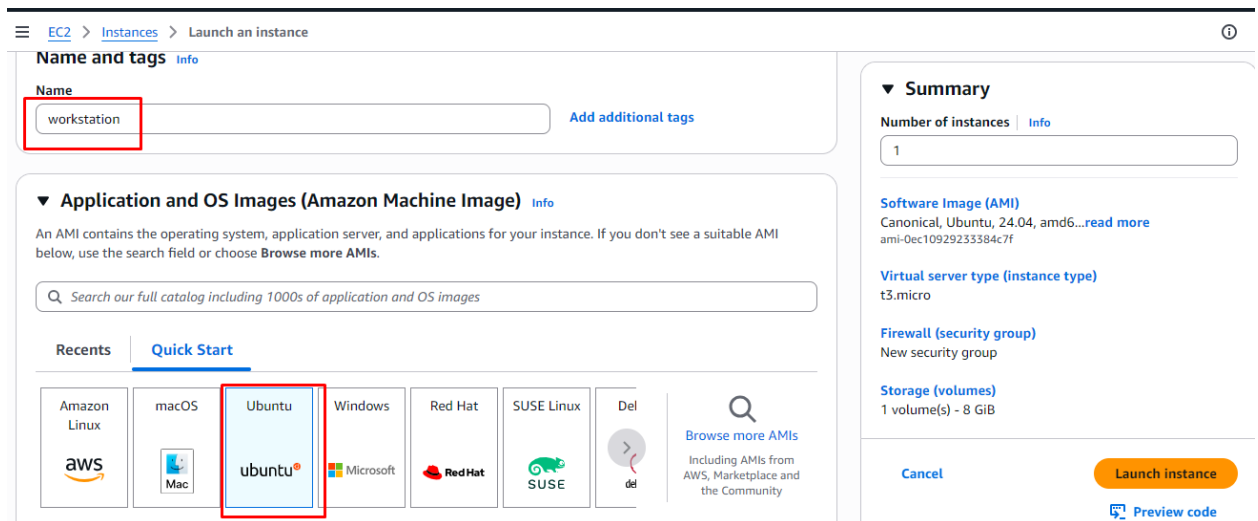
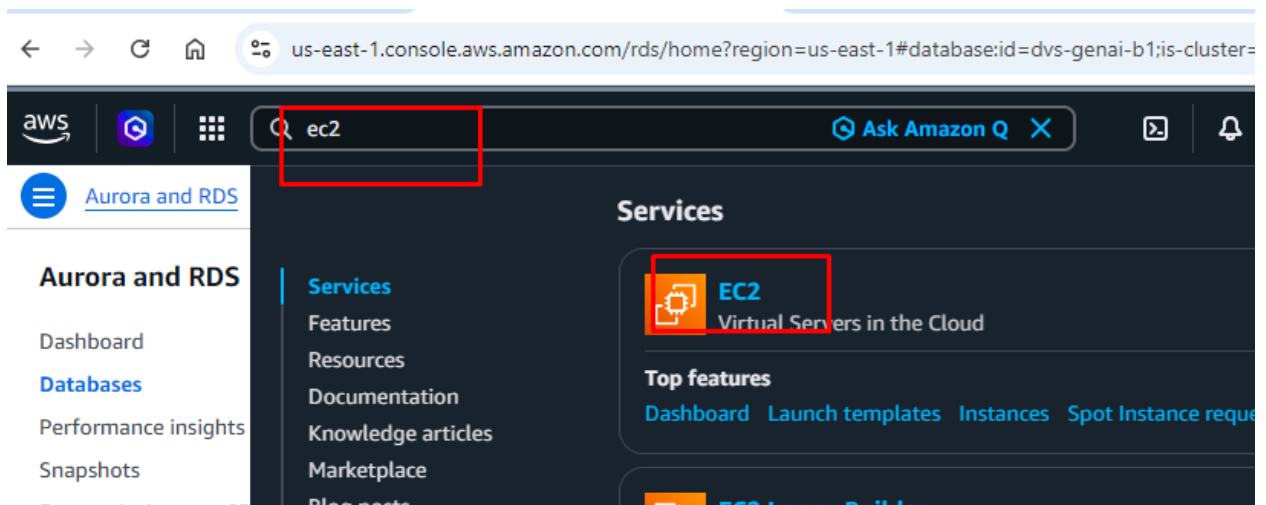
Port
3306

Endpoint type
Instance endpoint

▼ Additional configurations

Connectivity & security	Networking	Security
Endpoint & port Endpoint dvs-genai-b1.cqn6ou022fgc.us-east-1.rds.amazonaws.com Port 3306	Availability Zone us-east-1f VPC vpc-0b8fbad2d38436ee9 Subnet group	VPC security groups default (sg-08eca5fae8bedd3f5) Active Publicly accessible Yes

2.Create one EC2 Instance



EC2 > Instances > Launch an instance

GENAI-B1-KEY1

Create new key pair

▼ Network settings Info

Edit

Network Info

vpc-0b8fbad2d38436ee9

Subnet Info

No preference (Default subnet in any availability zone)

Auto-assign public IP Info

Enable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups Info

Select security groups

default sg-08eca5fae8bedd3f5

Compare security group rules

▼ Summary

Number of instances Info

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more

ami-0ec10929233384c7f

Virtual server type (instance type)

t3.micro

Firewall (security group)

default

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

Create a new RDS Database

Learn more

lancer

1, network gateway or Balancer

lancer

Create AWS budget

AWS Budgets allows you to create budgets, forecast spend, and take action on your costs and usage from a single location.

Create AWS budget

Manage CloudWatch alarms

Create or update Amazon CloudWatch alarms for the instance.

Manage CloudWatch alarms

View all instances

EC2 > Instances

Instances (1/1) Info

Connect

Instance state

Actions

Launch inst

Find Instance by attribute or tag (case-sensitive)

All states

☒	Name	Instance ID	Instance state	Instance type	Status check
☒	workstation	i-0a074e64d745f73fe	Running	t3.micro	Initializing

3.Login to RDS instance

sudo su - root

```
ubuntu@ip-172-31-26-137:~$ sudo su - root
root@ip-172-31-26-137:~# whoami
root
root@ip-172-31-26-137:~#
```

i-0a074e64d745f73fe (workstation)

sudo apt update

sudo apt install mysql-client

```
to locate package mysql
2-31-26-137:~# sudo apt update
::us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
::us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates
::us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backpor
::us-east-1.ec2.archive.ubuntu.com/ubuntu noble/univers
::security.ubuntu.com/ubuntu noble-security InRelease [
::us-east-1.ec2.archive.ubuntu.com/ubuntu noble/univers
::us-east-1.ec2.archive.ubuntu.com/ubuntu noble/univers
::us-east-1.ec2.archive.ubuntu.com/ubuntu noble/univers
::us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiv
Unchanged 40.8 MB in 6s (6333 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Some packages can be upgraded. Run 'apt list --upgradable' to see them.
root@ip-172-31-26-137:~# sudo apt install mysql-client
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
mysql-client-8.0 mysql-client-core-8.0 mysql-common
The following NEW packages will be installed:
```

```
0 upgraded, 4 newly installed, 0 to remove and 11 not upgraded.
Need to get 2779 kB of archives.
After this operation, 61.8 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 mysql-co
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host
root@ip-172-31-26-137:~# mysql --version
mysql Ver 8.0.45-0ubuntu0.24.04.1 for Linux on x86_64 ((Ubuntu))
root@ip-172-31-26-137:~#
```

Capture rds instance details

The screenshot shows the AWS Management Console for an Amazon RDS instance named 'dvs-genai-b1'. The instance is of type 'mysql'. The 'Endpoints' tab is selected, showing the 'Endpoint type' as 'Instance endpoint'. The 'Additional configurations' section is expanded, showing 'Connectivity & security' details. The 'Endpoint & port' section shows the 'Endpoint' as 'dvs-genai-b1.cqn6ou022fgc.us-east-1.rds.amazonaws.com' and the 'Port' as '3306'. The 'Networking' section shows the 'Availability Zone' as 'us-east-1f' and the 'VPC' as 'vpc-0b8fbad2d38436ee9'. The 'Security' section shows the 'VPC security groups' as 'default (sg-08eca5fae8bedd3f5)' and 'Active'.

```
mysql -u admin -padmin1234 -h dvs-genai-b1.cqn6ou022fgc.us-east-1.rds.amazonaws.com -P 3306
```

--u – username, -p – password, -P port no, -h – hostname

```
mysql Ver 8.0.45-0ubuntu0.24.04.1 for Linux on x86_64 ((Ubuntu))
root@ip-172-31-26-137:~# mysql -u admin -padmin1234 -h dvs-genai-b1.cqn6ou022fgc.us-east-1.rds.amazonaws.com -P 3306
mysql: [Warning] Using a password on the command line interface can be insecure.
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 29
Server version: 8.4.7 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
mysql>
```

show databases;

```
mysql> show databases;
+-----+
| Database          |
+-----+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+-----+
4 rows in set (0.01 sec)
```

```
n for the right syntax to use near '-genai-db' at line 1
mysql> create database dvsgenaidb;
Query OK, 1 row affected (0.00 sec)

mysql> show databases;
+-----+
| Database          |
+-----+
| dvsgenaidb         |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+-----+
5 rows in set (0.01 sec)
```

```
mysql> exit;
Bye
root@ip-172-31-26-137:~#
```

i-0a074e64d745f73fe (workstation)

Note: Make sure that you are deleting RDS instance at any cost ...

ec2

Ask Amazon Q

United States (N. Virginia)

ady_aws (4720-0804-6327)

shaan

DS > Databases > dvs-genai-b1

DS

<

jhts

n S3

ips

is

✔ Successfully created database **dvs-genai-b1**

You can use settings from dvs-genai-b1 to simplify configuration of sug while we finish creating your DB for you.

View connection details

Stop temporarily

Reboot

Delete

Set up EC2 connection

Set up Lambda connection

Migrate data from EC2 database

Create read replica

dvs-genai-b1

Modify

Actions ▲

Summary

DB identifier

Status

Delete dvs-genai-b1 instance



Permanently delete **dvs-genai-b1** DB instance. You can't undo this action.

 Proceeding with this action will delete the instance with all its content and can affect related resources. [Learn more](#) 

☐ Create final snapshot

Determines whether a final DB Snapshot is created before the DB instance is deleted.

☒ I acknowledge that upon instance deletion, automated backups, including system snapshots and point-in-time recovery, will no longer be available.

To avoid accidental deletion provide additional written consent.

To confirm deletion, type *delete me* into the field.

delete me

 We strongly recommend taking a final snapshot before instance deletion since after your instance is deleted, automated backups will no longer be available.

Cancel

Delete